

Smart Garage Gate – Future Mall

System Diagram

Inputs

- Entry Button → Represents a car entering the garage.
- Exit Button → Represents a car leaving the garage.

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Processing

ESP32

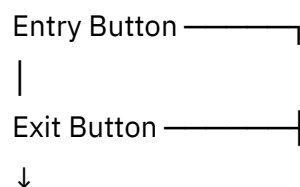
The ESP32 receives the signals from the Entry and Exit Buttons, counts the number of cars inside the garage, and controls the LEDs according to the garage status.

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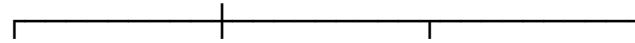
Outputs

- **Green LED** → Indicates that there are available parking spaces.
- **Red LED** → Indicates that the garage is Full.
- **Yellow LED** → Blinks once when a car enters successfully.
- **Blue LED** → Blinks once when a car exits successfully.

System Flow



ESP32



Green

LED



Red

LED



Yellow

LED



Blue

LED

Inputs → Processing → Outputs